

Introduction to IT System Suggestion (2024-2025)

PDF

For 2nd Semester of Polytechnic

Top 200 Questions with
Complete Answer
From All Chapters



Prepared by **NatiTute** YouTube Channel

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1. Which of the following not a browser?

a) Safari b) Android c) Chrome d) Edge

Explanation: Safari, Chrome, and Edge are all web browsers developed by Apple, Google, and Microsoft respectively. Android, on the other hand, is an operating system primarily used on mobile devices, not a web browser.

Answer: (b)

2. A web browser is an application that is used to- (a) find and retrieve webpage. (b) send e-Mail (c) prepare a document (d) none of these.

Explanation: A web browser (e.g., Chrome, Firefox) is an application that helps users access and display webpages from the internet.

Answer: (a)

3. In any search engine user can- (a) use of two or more keywords (b) use of + (c) use of And (d) All of these.

Explanation: Search engines allow users to refine their searches using multiple keywords, the + operator, and logical operators like AND to narrow down results.

Answer: (d)

4. Minimum number of bits required to represent 15 different characters individually: a) 15 b) 4 c) 5 d) 3

Explanation:

Number of bits $\geq \log_2(\text{Number of characters})$

Number of bits $\geq \log_2(15) \approx 3.91$

Since the number of bits must be an integer, we round up to the next whole number, which is 4.

Answer: (b)

5. Which of the following memory is used to design BIOS? a) RAM b) ROM c) Cache d) Virtual Memory.

Explanation: BIOS (Basic Input/Output System) is firmware stored in ROM because it needs to retain its data even when the computer is powered off.

Answer: (b)

6. A hard disc is divided into tracks, which are further sub divided into- a) Clusters b) Sectors c) Vectors d) Heads.

Explanation: A hard disk is divided into tracks, which are concentric circles on the disk. Each track is further divided into smaller units called sectors, which are the smallest storage units on the disk, typically holding 512 bytes of data.

Answer: (b)

7. Logical operations are performed in computer using- a) Memory Unit b) ALU c) CD d) ROM.

Explanation: The Arithmetic Logic Unit (ALU) is responsible for performing logical operations (such as AND, OR, NOT, XOR) and arithmetic operations (such as addition, subtraction, multiplication, and division) in a computer.

Answer: (b)

8. Equivalent Binary representation of the Gray code 0111 is- a) 0101 b) 1000 c) 1010 d) 1111

Explanation: When converting Gray code to binary, you compare each bit with the previous bit; if they are the same, copy a "0", if different, copy a "1". Following this rule for "0111", the binary equivalent is "0101".

Answer: (a)

9. DNS is- a) Horizontal naming system b) Vertical naming system c) client server system d) None of these.

Explanation: DNS (Domain Name System) follows a client-server architecture, where DNS clients (resolvers) request domain name resolution from DNS servers, which respond with the corresponding IP addresses.

Answer: (c)

10. Type of software used to create slideshow is -

a) Web design software

b) Presentation software

c) Word-processing software

d) Spreadsheet software

Explanation: Presentation software is specifically designed to create slideshows that are often used for visual presentations. Examples include Microsoft PowerPoint, Google Slides, and Keynote.

Answer: (b)

11. A simple tool that helps you create, organize, and analyze data. a) Browser b) Web viewer c) Spreadsheet d) Web server

Explanation: A spreadsheet is a tool used for organizing, analyzing, and storing data in tabular form. It allows users to perform calculations, create charts, and manipulate data efficiently, making it ideal for data analysis and organization.

Answer: (c)

12. The result of the expression $29+1=$ a) 11101

b) 11110 c) 11000 d) 10001

Explanation: Calculate the decimal result: $29 + 1 = 30$

Convert 30 to binary:

$$\frac{30}{2} = 15 \rightarrow 0 \text{ (LSB)}$$

$$\frac{15}{2} = 7 \rightarrow 1$$

$$\frac{7}{2} = 3 \rightarrow 1$$

$$\frac{3}{2} = 1 \rightarrow 1$$

$$\frac{1}{2} = 0 \rightarrow 1 \text{ (MSB)}$$

$\therefore 30$ in binary is 11110.

Answer: (b)

13. Decimal equivalent of 16 in hexadecimal (a) 22_{10} (b)

16_{10} (c) 10_{10} (d) 20_{10}

Explanation: $1 \times 16^1 + 6 \times 16^0 = 16 + 6 = 22$

Answer: (a)

14. The number of bits used to store a BCD digit is- (a) 2

(b) 1 (c) 4 (d) 8

Explanation: BCD (Binary-Coded Decimal) represents decimal numbers using 4 bits per digit.

Answer: (c)

15. What is the resultant binary equivalent of the decimal problem $49 + 1 =$ (a) 00110010 (b)

01010101 (c) 00110001 (d) 00110101

Explanation: $49 + 1 = 50$

$$50 / 2 = 25, \text{ remainder } 0$$

$$25 / 2 = 12, \text{ remainder } 1$$

$$12 / 2 = 6, \text{ remainder } 0$$

$$6 / 2 = 3, \text{ remainder } 0$$

$$3 / 2 = 1, \text{ remainder } 1$$

$$1 / 2 = 0, \text{ remainder } 1$$

Reading the remainders from bottom to top, we get the binary equivalent: 110010

Answer: (a)

16. Portion of information processing cycle in which computer acquires data from some source is called -

a) Input b) Output c) Processing d) Storage.

Explanation: Input is the stage in the information processing cycle where the computer acquires data from an external source, such as a keyboard, mouse, scanner, or other input devices. This data is then processed by the computer.

Answer: (a)

17. Binary equivalent of 97_{10} is- a) 1100000 b) 1100001

c) 1100010 d) 1100011

Solution:

Division	Quotient	Remainder
$97 \div 2$	48	1
$48 \div 2$	24	0
$24 \div 2$	12	0
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Reading from bottom to top: $(97)_{10} = (1100001)_2$

Answer: (b)

18. Example of Universal Gate-

a) NAND b) XOR c) OR d) AND

Explanation: A NAND gate is considered a universal gate because you can use it to construct any other type of logic gate, such as AND, OR, NOT, and even XOR gates, by combining multiple NAND gates appropriately.

Answer: (a)

19. The expression $Y = (A+B)(B+C)(C+A)$ shows the operation. a) AND b) POS c) SOP d) NAND

Explanation: POS (Product of Sums) means the expression consists of multiple sums (OR operations) that are multiplied (AND operation).

Answer: (b)

20. Which of the following is used for connecting computers in a large area such as country, continent etc

a) LAN b) WAN c) MAN d) PAN

Explanation: A WAN, or Wide Area Network, is designed to cover broad geographic areas such as countries or continents, connecting computers over long distances. In contrast, a LAN (Local Area Network) is limited to smaller areas like homes or offices, a MAN (Metropolitan Area Network) covers a city, and a PAN (Personal Area Network) is intended for personal devices usually within a short range.

Answer: (b)

21. Smart phone is an example of -

- a) Personal computer
- b) Analog computer
- c) Mainframe computer
- d) Super computer

Explanation: A smart phone is considered a type of personal computer (PC) because it is designed for individual use. It has computing capabilities similar to a traditional PC, including internet access, applications, and storage. Although it is much smaller and portable, it performs many tasks that a regular computer can.

22. Top listed super computer is- a) Sunway taihulight b) Sequoia c) Cori d) Mira

Explanation: Based on the latest TOP500 rankings as of November 2024, the top-listed supercomputer is El Capitan. Among the options provided, Sunway TaihuLight is the highest-ranked, previously holding the number one position in June 2016.

Answer: (a)

23. _____ is a type of software that is embedded in hardware.

Explanation: Firmware is a type of software embedded in hardware, providing low-level control for the device's functions (e.g., BIOS, microcontroller software).

Answer: Firmware

24. Universal gates are those gates that can be used for implementing any _____ function without any other types of gates.

Explanation: Universal gates (NAND and NOR) can be used to implement any Boolean function without needing other types of logic gates.

Answer: Boolean

25. $A+0=A$ is an example of _____ Law.

Explanation: $A + 0 = A$ is an example of the Identity Law in Boolean algebra, which states that adding zero to a variable does not change its value.

Answer: Identity Law

26. _____ is a batch file stored in the root directory of the hard-disk.

Explanation: AUTOEXEC.BAT is a batch file stored in the root directory of the hard disk in older Windows/DOS systems, executing commands at startup.

Answer: AUTOEXEC.BAT

27. A group of four bits is called a _____.

Explanation: A nibble is a group of four bits (e.g., 1010 is a nibble).

Answer: nibble

28. The advantage of DRAM over SRAM is that it is _____, can hold more data per chip and generate less heat.

Explanation: DRAM (Dynamic RAM) is cheaper than SRAM (Static RAM), holds more data per chip, and generates less heat, but it requires periodic refreshing.

Answer: cheaper

29. _____ software is machine independent.

Explanation: Application software is machine-independent, meaning it can run on different hardware platforms with compatible operating systems.

Answer: Application software

30. _____ is a service that automatically converts domain names into IP addresses.

Explanation: DNS is a service that converts domain names (e.g., google.com) into IP addresses to locate web resources.

Answer: DNS (Domain Name System)

31. Gray code equivalent of 1001 in Binary code is _____.

Explanation: The most significant bit (MSB) of the Gray code is the same as the MSB of the binary number. Each subsequent Gray code bit is obtained by performing an XOR operation between the corresponding binary bit and the previous binary bit.

Answer: 1101

32. What is the maximum number of outputs a logic gate can have?

Answer: The maximum number of outputs a logic gate can have is one. Logic gates process input signals and produce a single output based on the logic function they perform.

33. Write De Morgan's first Law.

Answer: De Morgan's First Law states:

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

This means that the complement of the OR operation of two variables is equal to the AND operation of their complements.

34. Identify the device which receives analog data and creates its digital equivalent.

Answer: The device that receives analog data and creates its digital equivalent is called an Analog-to-Digital Converter (ADC).

35. Define Booting.

Answer: Booting is the process of starting a computer and loading the operating system into memory so that the system becomes operational.

36. What is the first page of a website called?

Answer: The first page of a website is called the Home Page.

37. Write the full form of HTTP.

Answer: The full form of HTTP is Hypertext Transfer Protocol.

38. Write the full form of SSD.

Answer: The full form of SSD is Solid State Drive.

39. Differentiate between Data and Information.

Answer: Data refers to raw, unprocessed facts, figures, or symbols that lack meaning on their own, such as numbers, letters, or binary code. Information, on the other hand, is processed, organized, or structured data that provides meaning and context, making it useful for decision-making. For example, a list of test scores is data, but calculating the average score to assess student performance turns it into information.

40. What is BIOS?

Answer: BIOS (Basic Input/Output System) is firmware that initializes hardware components and starts the operating system during the booting process.

41. Write the utility of Cache Memory.

Answer: Cache Memory stores frequently accessed data and instructions to speed up processing and improve system performance.

42. Define Truth Table.

Answer: A Truth Table is a table that represents all possible input values and their corresponding output values for a logic gate or Boolean expression.

43. Write the utility of Command Line Interpreter.

Answer: A Command Line Interpreter (CLI) allows users to interact with a computer system by executing commands, typically through a text-based interface like Command Prompt or Terminal.

44. Write the full form of URL.

Answer: The full form of URL is Uniform Resource Locator.

45. Write down the full form of e-mail and www.

Answer:

E-mail: Electronic Mail

WWW: World Wide Web

46. Write down the objectives of Digital India.

Answer: The Digital India initiative aims to transform India into a digitally empowered society and knowledge economy. Its key objectives include:

1. **Digital Infrastructure Development:** Providing high-speed internet and digital services across the country.
2. **E-Governance & Online Services:** Ensuring seamless government services through digital platforms.
3. **Digital Literacy:** Promoting awareness and training citizens in digital skills.
4. **Cashless Economy:** Encouraging digital payments and financial inclusion.
5. **Job Creation & Innovation:** Supporting startups, IT industries, and digital entrepreneurship.
6. **Cybersecurity & Data Protection:** Ensuring safe and secure digital transactions.
7. **Smart Cities & Digital Connectivity:** Enhancing urban and rural digital services for better governance.

47. What is Booting?

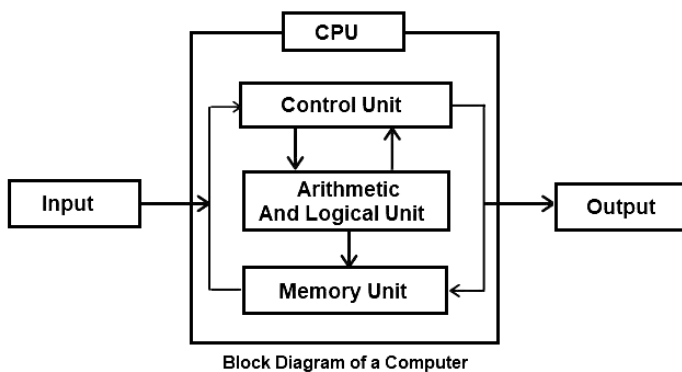
Answer:

Booting: Booting is the process of starting a computer and loading the operating system (OS) into memory.

There are two types of booting: Cold Booting, which starts the system from a powered-off state, and Warm Booting, which restarts the system without turning off the power. Once the OS is loaded, essential drivers and user settings are initialized, making the computer ready for use.

48. Draw the block diagram of a digital computer specifying all parts of it.

Answer:



Explanation of the Components:

- 1) **Input Devices:** Devices used to enter data (e.g., keyboard, mouse).
- 2) **CPU (Central Processing Unit)** – The brain of the computer:
 - **Control Unit (CU)** – Directs operations and manages data flow.
 - **Arithmetic Logic Unit (ALU)** – Performs mathematical and logical operations.
- 3) **Memory Unit** – Stores data temporarily and permanently:
 - **RAM (Random Access Memory)** – Temporary memory for active processes.
 - **ROM (Read-Only Memory)** – Permanent memory storing essential firmware.
- 4) **Storage Devices** – Used for long-term data storage (HDD, SSD, USB).
- 5) **Output Devices** – Displays or produces results (monitor, printer).

49. Why we use web browser?

Answer: A web browser is used to access, retrieve, and display information from the Internet. It allows users to visit websites, search for information, watch videos, download files, and interact with online services. Browsers interpret HTML, CSS, and JavaScript to render web pages and support security features like encryption and authentication for safe browsing. Examples include Google Chrome, Mozilla Firefox, and Microsoft Edge.

50. What is CPU and I/O bound jobs?

Answer: CPU-bound and I/O-bound jobs are two categories of tasks based on how they utilize system resources.

1) CPU-Bound Jobs:

- These tasks spend most of their time using the Central Processing Unit (CPU) for computations and processing.
- They require significant processing power and perform complex calculations or data manipulations.
- Examples include scientific simulations, mathematical calculations, and video rendering.

2) I/O-Bound Jobs:

- These tasks spend most of their time waiting for Input/Output (I/O) operations to complete, such as reading or writing data from disks, networks, or peripherals.
- They rely heavily on external devices and may cause the CPU to remain idle while waiting for data.
- Examples include file transfers, database queries, and printing documents.

51. Define ROM, PROM, EPROM and EEPROM.

Answer: The definitions of ROM, PROM, EPROM, and EEPROM are-

1) ROM (Read-Only Memory)

- A type of non-volatile memory that stores permanent data and instructions essential for booting the computer.
- Data in ROM cannot be modified or erased easily.
- Example: BIOS firmware in computers.

2) PROM (Programmable Read-Only Memory)

- A type of ROM that is initially blank and can be programmed once using special hardware.
- Once programmed, the data cannot be changed or erased.
- Used in applications where permanent storage of specific instructions is needed.

3) EPROM (Erasable Programmable Read-Only Memory)

- A type of PROM that can be erased using **ultraviolet (UV) light** and reprogrammed.
- It has a transparent window on top for UV light exposure.
- Used in firmware development before finalizing the permanent ROM.

4) EEPROM (Electrically Erasable Programmable Read-Only Memory)

- A type of ROM that can be erased and rewritten **electrically** multiple times.
- Unlike EPROM, it does not require UV light for erasure.
- Used in BIOS, microcontrollers, and modern storage devices like SSDs.

52. Why are search engines important?

Answer: Search engines are important because they help users quickly find relevant information on the internet by indexing and organizing vast amounts of data. They use algorithms to rank web pages based on keywords, relevance, and credibility, making it easier to access websites, articles, images, and videos. Popular search engines like Google, Bing, and Yahoo enhance productivity, support research, and provide accurate results in seconds.

53. State and prove the De-Morgan's Law in Boolean algebra.

Answer:

De Morgan's Laws in Boolean Algebra: De Morgan's Theorems provide an equivalence between AND, OR, and NOT operations. They state:

First Law: The complement of a product is equal to the sum of the complements.

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

Proof:

A	B	A.B	$\overline{A \cdot B}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

Since $\overline{A \cdot B}$ and $\overline{A} + \overline{B}$ produce the same output, 1st Law is proved.

Second Law: The complement of a sum is equal to the product of the complements.

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

Proof:

A	B	A + B	$\overline{A + B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Since $\overline{A + B}$ and $\overline{A} \cdot \overline{B}$ produce the same output, 2nd Law is proved.

54. Represent (5)₁₀ using 2-4-2-1 BCD code.

Answer: Finding the combination for 5:

We need to find a combination of these weights that adds up to 5.

The combination that works is:

$$1 \times 2 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 2 + 0 + 2 + 1 = 5$$

This correlates to the binary digits of 1011.

Therefore, the 2-4-2-1 BCD representation of 5 is 1011.

55. Convert Binary to Gray Code: (1101101)₂ = (?)_G

Answer:

Binary: 1 1 0 1 1 0 1

MSB (1): The MSB is directly copied to the Gray code.

$$1 \text{ XOR } 1 = 0$$

$$1 \text{ XOR } 0 = 1$$

$$0 \text{ XOR } 1 = 1$$

$$1 \text{ XOR } 1 = 0$$

$$1 \text{ XOR } 0 = 1$$

$$0 \text{ XOR } 1 = 1$$

Gray Code: 1 0 1 1 0 1 1 (Ans)

56. Express the following number in Gray code form:

(i) 15 (ii) 27

Solution:

(i) Convert 15 to binary:

$$15/2 \rightarrow 7 \text{ remainder } 1$$

$$7/2 \rightarrow 3 \text{ remainder } 1$$

$$3/2 \rightarrow 1 \text{ remainder } 1$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (15)_{10} = (1111)_2$$

Apply the Gray code conversion:

- First bit = 1 (same as binary)

- $1 \oplus 1 = 0$

- $1 \oplus 1 = 0$

- $1 \oplus 1 = 0$

$$\therefore \text{Gray Code for 15} = 1000 \text{ (Ans)}$$

(ii) Convert 27 to binary:

$$27/2 \rightarrow 13 \text{ remainder } 1$$

$$13/2 \rightarrow 6 \text{ remainder } 1$$

$$6/2 \rightarrow 3 \text{ remainder } 0$$

$$3/2 \rightarrow 1 \text{ remainder } 1$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (27)_{10} = (11011)_2$$

Apply the Gray code conversion:

- First bit = 1 (same as binary)

- $1 \oplus 1 = 0$

- $0 \oplus 1 = 1$

- $1 \oplus 1 = 0$

- $1 \oplus 1 = 0$

$$\therefore \text{Gray Code for 27} = 10100 \text{ (Ans)}$$

57. Convert $(10110)_2$ to its equivalent gray code.

Answer:

Steps to Convert Binary to Gray Code:

1. The first bit of the Gray code is the same as the first bit of the binary number.
2. Each subsequent Gray code bit is found by XORing the previous binary bit with the current binary bit:

Binary to Gray Code Conversion for $(10110)_2$

Binary (B)	1	0	1	1	0
Gray (G)	1	1	1	0	1

∴ Gray Code Output: 11101 (Ans)

58. Define universal gates. Why those gates are termed as universal gates?

Answer:

Universal Gates: Universal gates are logic gates that can be used to implement any Boolean function without needing other types of gates. The NAND and NOR gates are called universal gates.

NAND and NOR Called Universal Gates because-

1. **Functional Completeness:** Any Boolean function can be designed using only NAND or NOR gates.
2. **Replacement for Basic Gates:** These gates can replicate the functions of AND, OR, and NOT gates.
3. **Practicality in Circuit Design:** Using only one type of gate (NAND or NOR) simplifies hardware design and reduces manufacturing costs.

59. Represent $(-14)_{10}$ in 2's complement form using 8-bits representation.

Answer:

Step 1: Convert 14 to Binary (Positive Form)

Division	Quotient	Remainder
$14 \div 2$	7	0
$7 \div 2$	3	1
$3 \div 2$	1	1
$1 \div 2$	0	1

Reading the remainders from bottom to top

$$(14)_{10} = (1110)_2$$

Pad with leading zeros to make it 8 bits: 00001110

To get the negative representation,

Invert the bits (1's complement): 11110001

Add 1 to get the 2's complement:

$$11110001 + 1 = 11110010$$

$$\therefore (-14)_{10} = (11110010)_2$$

(in 8-bit 2's complement)

60. Convert the following as indicated:

i) $(175.25)_{10} = (?)_8$

ii) $(1011.101)_2 = (?)_{10}$

Solution:

i) Convert $(175.25)_{10}$ to Octal:

Step 1: Convert Integer Part (175) to Octal

Divide 175 by 8 and record the remainders:

Division	Quotient	Remainder
$175 \div 8$	21	7
$21 \div 8$	2	5
$2 \div 8$	0	2

Reading from bottom to top: $(175)_{10} = (257)_8$

Step 2: Convert Fractional Part (0.25) to Octal

Multiply 0.25 by 8 repeatedly:

$$0.25 \times 8 = 2.0$$

Since we get a whole number (2), the fractional part stops here: $(0.25)_{10} = (0.2)_8$.

$$\text{Final Answer: } (175.25)_{10} = (257.2)_8$$

ii) Convert $(1011.101)_2$ to Decimal:

We use positional values of binary:

$$(1011.101)_2$$

Step 1: Convert Integer Part (1011)

$$= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 11$$

Step 2: Convert Fractional Part (.101)

$$= 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= \frac{1}{2} + 0 + \frac{1}{8} = 0.5 + 0.125 = 0.625$$

$$\text{Final Answer: } (1011.101)_2 = (11.625)_{10}$$

61. How many type of Software? Give some example.

Answer: Software can be broadly classified into two main types:

1. **System Software:** System software helps run and manage computer hardware.

Examples:

Operating Systems: Windows, macOS, Linux, Android

Utility Software: Antivirus (Norton, McAfee), Disk Cleanup, File Manager

Device Drivers: Printer Driver, Graphics Driver

2. **Application Software:** Application software is designed for end-users to perform specific tasks.

Examples:

Productivity Software: Microsoft Office, Google Docs

Multimedia Software: Adobe Photoshop, VLC Media Player

Web Browsers: Google Chrome, Mozilla Firefox

Games: PUBG, Minecraft

62. What is meant by a cache memory? Explain.

Answer:

Cache Memory: Cache memory is a small, high-speed memory located close to the CPU that stores frequently accessed data and instructions. It helps improve processing speed by reducing the time the CPU takes to fetch data from the main memory (RAM).

How Cache Memory Works:

- (1) When the CPU needs data, it first checks the cache memory.
- (2) If the required data is found (cache hit), it is accessed quickly.
- (3) If the data is not found (cache miss), the CPU fetches it from RAM and stores a copy in the cache for future use.

Advantages of Cache Memory:

- (1) Increases CPU speed and efficiency.
- (2) Reduces dependency on slower RAM.
- (3) Enhances system performance in multitasking and gaming.

63. What are the characteristics of computers?

Answer:

Characteristics of Computers:

- (1) **Speed:** Computers process data and perform calculations at incredible speeds.
- (2) **Accuracy:** Computers follow instructions precisely, making them highly accurate.
- (3) **Automation:** Once programmed, a computer can execute tasks automatically without further human intervention.
- (4) **Versatility:** Computers can perform various tasks, from gaming to scientific research and data processing.
- (5) **Storage:** Computers have primary (RAM) and secondary (HDD/SSD) memory for storing data permanently or temporarily.
- (6) **Connectivity:** Modern computers can connect to the internet and communicate with other devices.
- (7) **Multitasking:** Computers can execute multiple programs or tasks simultaneously.
- (8) **Artificial Intelligence (AI) & Machine Learning:** Modern computers can learn from data and make decisions using AI.

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64. Prove: $\overline{A}B\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC = \overline{C}$

Answer:

L.H.S.

$$\begin{aligned} & \overline{A}B\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\ &= \overline{A}C(\overline{B} + B) + AC(\overline{B} + B) \\ &= \overline{A}C.1 + AC.1 \quad [\because \overline{B} + B = 1] \\ &= \overline{C}(\overline{A} + A) \\ &= \overline{C}.1 \quad [\because \overline{A} + A = 1] \\ &= \overline{C} \quad (\text{Proved}) \end{aligned}$$

65. Prove that: $AB + \overline{A}C + A\overline{B}C(AB + C) = 1$

Answer:

L.H.S.

$$\begin{aligned} & AB + \overline{A}C + A\overline{B}C(AB + C) \\ &= AB + \overline{A}C + A\overline{B}CAB + A\overline{B}CC \\ &= AB + \overline{A}C + A\overline{B}B CA + A\overline{B}C \quad [\because CC = C] \\ &= AB + \overline{A}C + 0 + A\overline{B}C \quad [\because \overline{B}B = 0] \\ &= AB + \overline{A} + \overline{C} + A\overline{B}C \quad [\because \overline{A}C = \overline{A} + \overline{C}] \\ &= \overline{A} + B + \overline{C} + A\overline{B} \\ &= \overline{A} + (\overline{B} + B) + \overline{C} \\ &= \overline{A} + 1 + \overline{C} = 1 = RHS \quad (\text{Proved}) \end{aligned}$$

66. Draw the logic diagram of an Ex-NOR gate and discuss its operation.

Answer: The Ex-NOR (Exclusive-NOR) gate is the complement of the Ex-OR (Exclusive-OR) gate. It outputs high (1) when both inputs are the same (either both 0 or both 1) and outputs low (0) when the inputs are different.

Logic diagram of an Ex-NOR gate:



Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Operation of XNOR Gate

When both inputs are 0:

XOR produces 0, and NOT inverts it to 1.

When one input is 1 and the other is 0:

XOR produces 1, and NOT inverts it to 0.

When both inputs are 1:

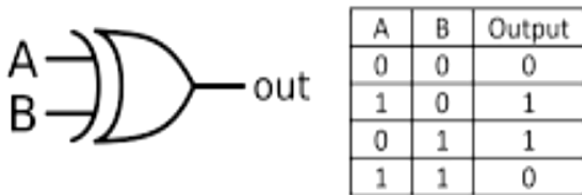
XOR produces 0, and NOT inverts it to 1.

Thus, XNOR outputs 1 when both inputs are the same (either both 0 or both 1) and 0 when they are different.

67. Draw the logic diagram of an Ex-OR gate and discuss its operation.

Answer:

Ex-OR: Logic diagram and truth table-



Operation of an Ex-OR Gate:

The Exclusive-OR (Ex-OR) gate has two inputs and one output. Its output is HIGH (1) if the inputs are different and LOW (0) if the inputs are the same.

The Boolean expression is:

$$Y = A \oplus B = \bar{A} \cdot B + A \cdot \bar{B}$$

This can be interpreted as follows:

- $\bar{A}B \rightarrow$ Output is 1 when A = 0 and B = 1
- $A\bar{B} \rightarrow$ Output is 1 when A = 1 and B = 0
- In all other cases, the output is 0.

68. Simplify the Boolean expression:

$$(X + Y)(X + \bar{Y})(\bar{X} + Z)$$

Answer:

$$(X + Y)(X + \bar{Y})(\bar{X} + Z)$$

[By absorption theorem $(X + Y)(X + \bar{Y}) = (X + Y\bar{Y})$]

$$= (X + Y\bar{Y})(\bar{X} + Z)$$

$$= (X + 0)(\bar{X} + Z)$$

$$= X\bar{X} + XZ$$

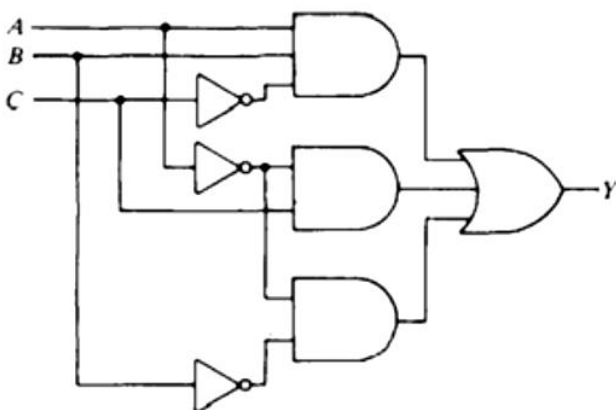
$$= 0 + XZ = XZ \quad (\text{Ans})$$

69. Draw the logic diagram of the Boolean expression:

$$Y = ABC + \bar{A}C + \bar{A}\bar{B}$$

Answer: Logic diagram of the Boolean expression:

$$Y = ABC + \bar{A}C + \bar{A}\bar{B}$$



70. Convert: $(27.23)_{10} = (?)_2$ and $(542)_8 = (?)_{16}$

Answer:

Convert $(27.23)_{10}$ to Binary :

Step 1: Divide 27 by 2, noting quotients and remainders:

$$27 \div 2 = 13, \text{ remainder } 1$$

$$13 \div 2 = 6, \text{ remainder } 1$$

$$6 \div 2 = 3, \text{ remainder } 0$$

$$3 \div 2 = 1, \text{ remainder } 1$$

$$1 \div 2 = 0, \text{ remainder } 1$$

Reading remainders from bottom to top: $(27)_{10} = (11011)_2$

Step 2: Convert the Fractional Part $(0.23)_{10} \rightarrow (?)_2$

Multiply by 2 and record integer parts:

$$0.23 \times 2 = 0.46 \rightarrow \text{Integer} = 0$$

$$0.46 \times 2 = 0.92 \rightarrow \text{Integer} = 0$$

$$0.92 \times 2 = 1.84 \rightarrow \text{Integer} = 1$$

$$0.84 \times 2 = 1.68 \rightarrow \text{Integer} = 1$$

$$0.68 \times 2 = 1.36 \rightarrow \text{Integer} = 1$$

(Stopping here with five decimal places)

$$\text{Thus, } (0.23)_{10} \approx (0.00111)_2$$

$$\therefore (27.23)_{10} \approx (11011.00111)_2 \quad (\text{Ans})$$

Convert $(542)_8$ to Hexadecimal (Base 16)

Step 1: Convert $(542)_8$ to Decimal (Base 10)

$$(542)_8 = 5 \times 8^2 + 4 \times 8^1 + 2 \times 8^0 = (354)_{10}$$

Step 2: Convert $(354)_{10}$ to Hexadecimal (Base 16)

$$1. \quad 354 \div 16 = 22, \text{ remainder } 2$$

$$2. \quad 22 \div 16 = 1, \text{ remainder } 6$$

$$3. \quad 1 \div 16 = 0, \text{ remainder } 1$$

Reading from bottom to top: $(354)_{10} = (162)_{16}$

$$\therefore (542)_8 = (162)_{16} \quad (\text{Ans})$$

71. State the difference between DRAM and SRAM.

Answer:

(b) Difference Between DRAM and SRAM

DRAM (Dynamic RAM)	SRAM (Static RAM)
Needs constant refreshing to maintain data.	Does not need refreshing, making it faster.
Slower (access time: 50–100 ns).	Faster (access time: 10–20 ns).
Consumes more power due to frequent refreshing.	Consumes less power since it holds data without refreshing.
Cheaper and widely used for main memory (RAM) in computers.	More expensive, used in cache memory (CPU cache).
Higher storage capacity but slower.	Lower storage capacity but much faster.

72. Write the main difference between Multitasking and Multiprogramming.

Answer:

Difference Between Multitasking and Multiprogramming:

Multitasking	Multiprogramming
Runs multiple tasks (processes) simultaneously by rapidly switching between them.	Runs multiple programs at the same time by storing them in memory.
CPU switches between tasks quickly, giving the illusion that all are running simultaneously.	CPU executes one program at a time, but multiple programs reside in memory.
Requires user interaction; common in modern operating systems (Windows, macOS, Linux).	Does not require user interaction; mostly used in batch processing.
More efficient CPU utilization by allowing multiple applications to run together.	Increases CPU utilization by loading multiple programs but executes one at a time.
Example: Running a web browser, music player, and text editor simultaneously.	Example: Running a compiler while another program is waiting for execution in memory.

73. State the difference between HDD and SSD memory

Answer:

(a) Difference Between HDD and SSD Memory

HDD (Hard Disk Drive)	SSD (Solid State Drive)
Uses spinning magnetic disks to store data.	Uses flash memory chips (no moving parts).
Slower read/write speeds due to mechanical parts.	Faster read/write speeds (10x or more than HDD).
More power consumption and generates heat.	Uses less power and stays cooler.
More storage capacity at a lower cost.	More expensive but faster and more durable.
Heavier and prone to mechanical failure.	Lighter, shock-resistant, and long-lasting.
Common in older computers and large storage devices.	Used in modern laptops, high-performance PCs, and gaming consoles.

74. Name the gates that are used as universal gates.

Answer:

Universal Gates: The two logic gates that are considered universal gates are:

1. NAND Gate (NOT AND)
2. NOR Gate (NOT OR)

These gates are called universal because they can be used to construct any other logic gate (AND, OR, NOT, XOR, XNOR) and even entire digital circuits.

75. What are ASCII and EBCDIC Codes?

Answer:

ASCII (American Standard Code for Information Interchange):

- ASCII is a 7-bit character encoding system that represents 128 characters (0 to 127).
- It includes English letters (A-Z, a-z), digits (0-9), punctuation marks, and control characters (like Enter and Backspace).
- Extended ASCII is an 8-bit version (0-255) used in modern systems.
- Used in: Windows, Unix, Linux, programming languages (C, Python, Java).

Example:

- 'A' = 65 in ASCII
- 'a' = 97 in ASCII
- '5' = 53 in ASCII

EBCDIC (Extended Binary Coded Decimal Interchange Code)

- EBCDIC is an 8-bit character encoding system developed by IBM, representing 256 characters.
- It is used mainly in IBM mainframe and midrange computers.
- Unlike ASCII, EBCDIC has different character sequences and is not widely used in modern systems.

Example:

- 'A' = 193 in EBCDIC
- 'a' = 129 in EBCDIC
- '5' = 245 in EBCDIC

76. Write the differences between Primary Memory and Secondary Memory.

Answer:

Differences between Primary Memory and Secondary Memory:

Primary Memory	Secondary Memory
It is used by the CPU to store data temporarily for quick access.	It is used for long-term data storage.
Very fast in speed (e.g., RAM operates in nanoseconds).	Slower compared to primary memory.
Volatile – data is lost when power is turned off.	Non-volatile – data remains stored even when power is off.
Examples: RAM, Cache Memory, Registers.	Examples: Hard Disk Drive (HDD), Solid-State Drive (SSD), USB drives.
Limited storage capacity (measured in GBs).	Large storage capacity (measured in GBs or TBs).
Used for processing and executing programs.	Used for storing data, files, and applications permanently.

77. Write the differences between internet browser and search engine.

Answer:

Differences Between Internet Browser and Search Engine:

Internet Browser	Search Engine
A software application used to access and display web pages.	A web-based tool used to search for information on the internet.
Loads and displays websites using URLs.	Helps users find websites, images, videos, and other content based on keywords.
Examples: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.	Examples: Google, Bing, Yahoo, DuckDuckGo.
Needed to open any website by entering its address (URL).	Used inside a browser to search for information.
Uses protocols like HTTP and HTTPS to fetch and render web pages.	Uses web crawlers to index and rank websites.

78. Why should we use more than one search engine simultaneously?

Answer: Different search engines have unique algorithms, databases, and indexing methods, so using multiple search engines can provide a wider range of results, uncovering more diverse and relevant information. It also helps verify the accuracy of information and find sources that may not appear on a single search engine.

79. How is 8-4-2-1 code different from 4-2-2-1 code?

Answer:

8-4-2-1 Code	4-2-2-1 Code
Uses weights 8, 4, 2, 1	Uses weights 4, 2, 2, 1
Follows standard Binary-Coded Decimal (BCD)	Uses a different weighting system
Example: Decimal 5 = 0101	Example: Decimal 5 = 1100
Widely used in digital systems	Less commonly used, specialized applications

80. "Two's complement is the most popular technique to represent signed number". Justify the statement.

Answer: Two's complement is widely used because it simplifies arithmetic operations, allowing addition and subtraction to be performed using the same binary rules as unsigned numbers. It also eliminates the problem of having both positive and negative zero, unlike sign-magnitude representation. Additionally, it is hardware-friendly and allows efficient calculations.

81. Is there any difference between compiler and an interpreter?

Answer: Yes, there is a difference between a compiler and an interpreter-

Compiler	Interpreter
Translates the entire source code before execution	Translates and executes code line by line
Faster execution after compilation	Slower execution due to line-by-line translation
Displays all errors after compilation	Stops execution at the first error
Used in languages like C, C++	Used in languages like Python, JavaScript

82. What are the meant by 1's and 2's complements of a binary number?

Answer: In binary arithmetic, 1's complement and 2's complement are methods used to represent negative numbers and perform subtraction.

1's Complement: It is obtained by flipping all the bits of a binary number (changing 0s to 1s and 1s to 0s).

Example:

Binary number: 1010

1's Complement: 0101

2's Complement: It is obtained by adding 1 to the 1's complement of a binary number.

Example:

Binary number: 1010

1's Complement: 0101

2's Complement: $0101 + 1 = 0110$

2's complement is widely used because it simplifies binary arithmetic operations like addition and subtraction.

83. How can two numbers be subtracted only through addition?

Answer: Two numbers can be subtracted using addition by utilizing the two's complement method. This technique allows subtraction to be performed as an addition operation in binary arithmetic.

Example: $7 - 5$ using 4-bit Binary

$$7 = 0111, 5 = 0101$$

Find two's complement of 5:

$$1\text{'s complement of } 0101 = 1010$$

$$\therefore 2\text{'s complement of } 0101 = 1010 + 1 = 1011$$

$$\therefore 7 - 5 = 0111 + 1011$$

$$= 0010 \text{ (taken 4 bit ignoring carry bit)}$$

$$= (2)_{10}$$

84. Add the following numbers using the 2's complement method:

(i) -48 and +31 (ii) -64 and +46

Solution:

(i) -48 and +31

$$48/2 \rightarrow 24 \text{ remainder } 0$$

$$24/2 \rightarrow 12 \text{ remainder } 0$$

$$12/2 \rightarrow 6 \text{ remainder } 0$$

$$6/2 \rightarrow 3 \text{ remainder } 0$$

$$3/2 \rightarrow 1 \text{ remainder } 1$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (48)_{10} = (00110000)_2 \text{ [taking 8 bit binary]}$$

$$1\text{'s complement of } 48 = 11001111$$

$$\therefore 2\text{'s complement} = -48 = 11001111 + 1 \\ = 11010000$$

$$31/2 \rightarrow 15 \text{ remainder } 1$$

$$15/2 \rightarrow 7 \text{ remainder } 1$$

$$7/2 \rightarrow 3 \text{ remainder } 1$$

$$3/2 \rightarrow 1 \text{ remainder } 1$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (31)_{10} = (00011111)_2$$

$$\therefore -48 + 31$$

$$= 11010000 + 00011111$$

$$= 11101111 \text{ (no overflow)}$$

It is a negative number (since the leftmost bit is 1).

$$1\text{'s complement: } 00010000$$

$$2\text{'s complement:}$$

$$= 00010000 + 1$$

$$= 00010001$$

$$= 1 \times 2^4 + 1 \times 2^0 = (17)_{10}$$

$$\text{Final answer: } (-17)$$

(ii) -64 and +46

$$64/2 \rightarrow 32 \text{ remainder } 0$$

$$32/2 \rightarrow 16 \text{ remainder } 0$$

$$16/2 \rightarrow 8 \text{ remainder } 0$$

$$8/2 \rightarrow 4 \text{ remainder } 0$$

$$4/2 \rightarrow 2 \text{ remainder } 0$$

$$2/2 \rightarrow 1 \text{ remainder } 0$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (64)_{10} = (01000000)_2 \text{ [taking 8 bit binary]}$$

$$1\text{'s complement of } 01000000 = 10111111$$

$$2\text{'s complement} = 10111111 + 1 = 11000000$$

$$46/2 \rightarrow 23 \text{ remainder } 0$$

$$23/2 \rightarrow 11 \text{ remainder } 1$$

$$11/2 \rightarrow 5 \text{ remainder } 1$$

$$5/2 \rightarrow 2 \text{ remainder } 1$$

$$2/2 \rightarrow 1 \text{ remainder } 0$$

$$1/2 \rightarrow 0 \text{ remainder } 1$$

$$\therefore (46)_{10} = (00101110)_2 \text{ [taking 8 bit binary]}$$

$$\therefore -64 + 46$$

$$= 11000000 + 00101110$$

$$= 11101110 \text{ (no overflow)}$$

It is a negative number (since the leftmost bit is 1).

$$1\text{'s complement} = 00010001$$

$$2\text{'s complement}$$

$$= 00010001 + 1$$

$$= 10010$$

$$= 1 \times 2^4 + 1 \times 2^1 = (18)_{10}$$

$$\text{Final answer} = (-18)$$

85. What do you mean by Universal Logic Gates? Why Universal Logic Gates are called Universal?

Answer:

Universal Gates: Universal gates are logic gates that can be used to implement any Boolean function without needing other types of gates. The NAND and NOR gates are called universal gates.

NAND and NOR Called Universal Gates because-

1. Functional Completeness: Any Boolean function can be designed using only NAND or NOR gates.

2. Replacement for Basic Gates: These gates can replicate the functions of AND, OR, and NOT gates.

3. Practicality in Circuit Design: Using only one type of gate (NAND or NOR) simplifies hardware design and reduces manufacturing costs.

86. Why SSD is faster than Hard Disk?

Answer: SSDs are faster than HDDs because they use flash memory instead of mechanical parts, allowing instant data access with no moving components. They have lower latency, faster read/write speeds (500 MB/s to several GB/s vs. 100-200 MB/s for HDDs), and better random access performance. SSDs also consume less power, reduce boot times, and eliminate fragmentation issues, making them significantly more efficient and responsive than HDDs.

87. Write the utility of Excess-3 code.

Answer: Excess-3 (XS-3) code is a self-complementary binary-coded decimal (BCD) system used in digital arithmetic and error detection. It simplifies subtraction by allowing easy computation of the 9's complement through bit inversion. Unlike standard BCD, Excess-3 eliminates invalid states, making it more reliable for digital circuit design. It is commonly used in calculators, digital clocks, and counters due to its ability to simplify arithmetic operations. Additionally, its structure helps in error detection and correction, making it useful in reliable computing systems.

88. What is a gray code? Why is it important?

Answer:

Gray Code: Gray code, also known as Reflected Binary Code (RBC), is a binary number system where two successive values differ by only one bit. This one-bit change minimizes errors during the transition from one value to another.

Importance of Gray Code:

- 1) **Error Reduction:** Since only one bit changes at a time, the risk of errors in digital circuits, especially in sensors and rotary encoders, is reduced.
- 2) **Data Integrity:** In mechanical systems, Gray code prevents misinterpretation of data due to noise or glitches during transitions.
- 3) **Efficient in Digital Systems:** It is used in analog-to-digital converters (ADCs) and digital position encoders for accurate reading.
- 4) **Simplifies Circuit Design:** Gray code reduces the complexity of error detection and correction circuits.

89. Convert the number $(5674)_8$ into the hexadecimal system.

Answer:

Step 1: Convert Octal (Base 8) to Binary (Base 2)
Each octal digit is represented by a 3-bit binary equivalent:

Octal Digit	Binary Equivalent
5	101
6	110
7	111
4	100

Thus, $(5674)_8 = 101\ 110\ 111\ 100_2$

Step 2: Convert Binary (Base 2) to Hexadecimal (Base 16)

Group the binary digits into 4-bit groups from right to left:

$101\ 110\ 111\ 100 \rightarrow 0001\ 0110\ 1111\ 0100$

(Adding a leading 0 to make complete 4-bit groups)

Now, convert each 4-bit group into hexadecimal:

Binary (4-bit group)	Hex Equivalent
0001	1
0110	6
1111	F
0100	4

$\therefore (5674)_8 = (16F4)_{16}$ (Ans)

{Prepared by [NatiTute](#) YouTube Channel}

1. Operating Systems used in the computer system for-

- a) File management b) Process management
c) Device management d) All of these

Explanation: An Operating System (OS) manages files, processes, and devices in a computer system. It handles file management by organizing and storing data, process management by controlling running applications, and device management by coordinating hardware like printers and storage devices.

Answer: (d)

2. Free and efficient operating system is ____.

- a) UNIX b) Windows c) MAC d) None of these.

Explanation: UNIX is a free and efficient operating system, widely used for its stability, security, and multitasking capabilities.

Answer: (a)

3. Which of the following is an example of a Real Time Operating System – a) MAC b) MS-DOS c) Windows 10 d) Process Control.

Explanation: A Real-Time Operating System (RTOS) is designed to manage hardware resources and run applications with strict timing constraints. It is often used in embedded systems, robotics, medical devices, and industrial automation.

Answer: (d)

4. Which of the following is not an operating system?- a) Windows b) Linux c) Oracle d) DOS.

Explanation: Oracle is not an operating system; it is a relational database management system (RDBMS) primarily used for managing and analyzing large amounts of data.

Answer: (c)

5. The Operating System _____ is responsible for interacting with hardware.

Explanation: The Kernel is the core part of an operating system responsible for interacting with hardware and managing system resources.

Answer: Kernel

6. In _____ scheduling, each process is given a small amount of time to execute.

Explanation: In Round-robin scheduling, each process is assigned a fixed time slot (time quantum) before moving to the next process in the queue.

Answer: Round-robin

7. What is operating system?

Answer:

Operating System: An Operating System (OS) is system software that acts as an interface between the computer hardware and the user. It manages hardware resources, provides essential services for application software, and ensures smooth communication between different programs and devices.

The primary functions of an operating system include:

- 1) **Process Management:** Controls and schedules processes, ensuring efficient execution.
- 2) **Memory Management:** Allocates and deallocates memory space as needed by programs.
- 3) **File Management:** Handles the creation, storage, retrieval, and management of data files.
- 4) **Device Management:** Coordinates and controls the use of hardware devices like printers, disks, and monitors.
- 5) **User Interface:** Provides a graphical or command-line interface for users to interact with the system.

Examples: Windows, macOS, Linux, and Android.

8. Write a short note on Types of OS.

Answer:

Types of Operating Systems

Operating Systems (OS) manage computer hardware and software resources. Based on their functionality and purpose, they can be classified into the following types:

- 1) **Batch Operating System**
 - Processes similar jobs in batches without user interaction.
 - Example: IBM OS/360.
- 2) **Time-Sharing Operating System**
 - Allows multiple users to access the system simultaneously by time-sharing resources.
 - Example: UNIX, Linux.
- 3) **Distributed Operating System**
 - Manages a group of interconnected computers and makes them appear as a single system.
 - Example: LOCUS, Windows Server.
- 4) **Real-Time Operating System (RTOS)**
 - Provides immediate response for time-critical tasks, often used in embedded systems.
 - Example: VxWorks, FreeRTOS.
- 5) **Network Operating System (NOS)**
 - Facilitates communication and resource sharing among computers in a network.
 - Example: Microsoft Windows Server, Linux Server.
- 6) **Mobile Operating System**
 - Designed for smartphones and tablets, offering touch interfaces and mobile app support.
 - Example: Android, iOS.

9. What is time sharing operating system?

Answer: A time-sharing operating system is a type of operating system that allows multiple users to access a computer system simultaneously by sharing the CPU time. It efficiently switches between different tasks so that users experience a continuous and interactive environment.

Example: UNIX, Linux (in multi-user mode)

10. Is it possible to run a computer without operating system? Write your opinion.

Answer: No, it is not practical to run a computer without an operating system because the OS manages hardware, software, memory, and user interactions. However, in specialized cases (such as embedded systems), computers can function without a conventional OS.

11. Give some importance functions of Operating System.

Answer:

Important Functions of an Operating System (OS):

An Operating System (OS) is essential software that manages computer hardware and software resources while providing services for applications. Its key functions include:

- 1) **Process Management:** The OS handles multiple processes, scheduling tasks efficiently for smooth execution.
- 2) **Memory Management:** It allocates and manages RAM, ensuring optimal use of memory resources.
- 3) **File Management:** The OS organizes, stores, retrieves, and secures files in storage devices.
- 4) **Device Management:** It controls hardware components like printers, keyboards, and USB devices using drivers.
- 5) **User Interface (UI):** Provides a graphical or command-line interface for user interaction.
- 6) **Security & Access Control:** Protects data through authentication, encryption, and permission settings.
- 7) **Networking:** Manages internet and network connections, enabling communication between devices.
- 8) **Error Detection & Handling:** Monitors system health, detecting and resolving errors or crashes.

12. Explain the advantages of multi-user and multi-tasking OS.

Answer:

Multi-User OS Advantages:

- 1) **Resource Sharing:** Multiple users can access shared resources like files, applications, and hardware (printers, storage) simultaneously.
- 2) **Cost Efficiency:** Reduces hardware costs as multiple users can work on a single system instead of requiring separate machines.
- 3) **Centralized Management:** Administrators can control user access, security, and data management efficiently.
- 4) **Data Consistency:** Ensures that all users have access to updated and synchronized data, reducing redundancy.
- 5) **Remote Accessibility:** Users can log in and work from different locations, enhancing collaboration and flexibility.

Multi-Tasking OS Advantages:

1. **Increased Productivity:** Users can run multiple applications at the same time, such as browsing the web while editing a document.
2. **Efficient CPU Utilization:** The OS allocates CPU time efficiently, ensuring smooth performance without wasting processing power.
3. **Better System Performance:** Reduces idle time by switching between tasks efficiently, improving overall speed.
4. **User Convenience:** Allows seamless switching between applications, making multitasking easier and improving workflow.
5. **Error Handling:** If one task crashes, the OS isolates it without affecting other running processes, improving system stability.

13. What is a bootstrap program in OS?

Answer: A bootstrap program in an operating system (OS) is a small, specialized piece of software responsible for initializing the system when a computer is powered on or restarted. It is also known as the bootloader.

Functions of a Bootstrap Program

- 1) **Hardware Initialization:** It performs basic hardware checks (Power-On Self Test - POST) and initializes essential components like memory, CPU, and input/output devices.
- 2) **Load the OS:** It locates the operating system from the storage (usually from the hard drive or SSD) and loads it into memory (RAM).
- 3) **Transfer Control:** After loading, it hands over control to the operating system kernel, which then takes charge of the system.

Example: BIOS (Basic Input/Output System)

14. Write the advantages of multiprocessing operating system.

Answer:

Advantages of a Multiprocessing Operating System:

- 1) **Increased Performance:** Multiple processors work simultaneously, improving the system's speed and efficiency.
- 2) **Better Resource Utilization:** CPU resources are optimally utilized, reducing idle time.
- 3) **Parallel Processing:** Tasks can be divided and executed concurrently, speeding up computations.
- 4) **Improved Reliability:** If one processor fails, others can continue functioning, increasing system stability.
- 5) **Faster Task Execution:** Multiple programs can run simultaneously without slowing down the system.
- 6) **Efficient Multitasking:** Users can perform multiple tasks smoothly without performance lag.
- 7) **Better System Throughput:** The number of tasks processed per unit time is higher compared to single-processor systems.

15. Write down the differences between system software and application software.

Answer:

Differences Between System Software and Application Software:

System Software	Application Software
Software that manages hardware and system resources.	Software designed for specific user tasks.
Provides a platform for other software to run.	Helps users perform specific tasks like writing, browsing, or gaming.
Operating Systems (Windows, Linux), Device Drivers, Utilities.	MS Word, Google Chrome, Photoshop, Games.
Works independently and supports application software.	Runs on system software and depends on it.
Minimal direct interaction, mostly works in the background.	Users interact directly to perform tasks.
Manages hardware, processes, memory, and system operations.	Used for productivity, entertainment, or communication.
Comes pre-installed with the system or is required for basic operation.	Installed as per user requirements.

16. Describe the general architecture of an operating system.

Answer: The general architecture of an operating system (OS) can be broken down into several key components:

Operating System Layers

1. **Firmware:** Low-level software that controls hardware components.
2. **Kernel:** Core part of the OS, responsible for managing hardware resources and providing basic services.
3. **Device Drivers:** Manage communication between the kernel and hardware devices.
4. **System Libraries:** Provide a interface for applications to interact with the kernel.
5. **System Services:** Manage tasks such as process scheduling, memory management, and file management.
6. **User Interface:** Provides a way for users to interact with the OS, such as command-line interfaces (CLI) or graphical user interfaces (GUI).

17. What is the difference between Batch Processing System and Real time Processing System?

Answer:

Difference Between Batch Processing System and Real-Time Processing System:

Batch Processing System	Real-Time Processing System
Processes data in groups (batches) without user interaction.	Processes data instantly with minimal delay.
Slower, as tasks are executed sequentially.	Fast, as tasks are executed in real-time.
No immediate interaction; jobs are processed later.	Requires immediate user or system interaction.
Examples: Payroll processing, bank statement generation, bill processing.	Examples: Air traffic control, online banking transactions, self-driving cars.
Suitable for large volumes of data that do not require immediate action.	Required for critical applications where instant results are needed.

18. What are the difference between multiprogramming and time sharing System?

Answer:

Differences between Multiprogramming and Time-Sharing System

Multiprogramming System	Time-Sharing System
Increases CPU utilization by running multiple programs at the same time.	Allows multiple users to access the system simultaneously.
The CPU switches between programs when one is waiting for I/O.	The CPU switches between tasks rapidly to provide each user with a short execution time (time slice).
Used in batch processing systems.	Used in interactive systems where multiple users work concurrently.
Example: Running multiple background processes in an operating system.	Example: Online banking systems where multiple users perform transactions at the same time.

19. What is Batch Operating System?

Answer: A Batch Operating System is an operating system that processes a series of jobs (batches) without user interaction. In this system, similar types of jobs are grouped together and executed sequentially, often during off-peak hours.

How It Works

- 1) Users prepare their jobs offline using punch cards, magnetic tapes, or disks.
- 2) The jobs are submitted to the computer operator.
- 3) The operator loads the jobs into the system as a batch.
- 4) The OS executes the batch jobs one by one without requiring further user input.
- 5) After execution, the results are provided to the users.

Characteristics of Batch OS

- 1) Jobs are processed in bulk.
- 2) Minimal user interaction during execution.
- 3) Efficient for tasks requiring large data processing.
- 4) Suitable for payroll processing, data analysis, and transaction processing.

Examples of Batch Operating Systems

- 1) IBM OS/360
- 2) UNIVAC

20. What do you mean by client server architecture Explain?

Answer:

Client-Server Architecture: Client-Server Architecture is a computing model that divides tasks between service providers (servers) and service requesters (clients). It is widely used in networks and applications where multiple users need to access resources, data, or services.

Components of Client-Server Architecture:

- 1) **Client**
 - The device or application that requests services from the server.
 - Examples: Web browsers, mobile apps, or computers.
- 2) **Server**
 - A powerful computer or software providing resources or services like data storage, files, or applications.
 - Examples: Web servers, database servers, and email servers.
- 3) **Network**
 - The medium that connects clients and servers, allowing communication.
 - Examples: LAN, WAN, or the internet.

How It Works:

- 1) The client sends a request to the server (e.g., requesting a web page).
- 2) The server processes the request, retrieves the necessary data, and sends a response back to the client.
- 3) The client displays or uses the received data.

21. What is a kernel?

Answer: The kernel is the core part of an OS that directly interacts with hardware and manages system resources.

22. What is a thread?

Answer: A thread is the smallest unit of a process that can be scheduled for execution.

23. What is virtual memory?

Answer: Virtual memory is a memory management technique that uses part of the storage (like a hard drive) to simulate extra RAM.

24. What is deadlock in an operating system?

Answer: Deadlock occurs when a set of processes are blocked because each process is waiting for a resource held by another process.

1. Which of the following is a pictorial representation of an algorithm?

- a) Pseudocode b) Program c) Flowchart d) None

Explanation: A flowchart is a pictorial representation of an algorithm using symbols to illustrate the steps and flow of a process. It visually represents the logic and sequence of operations.

Answer: (c)

2. A detailed flowchart is called ____.

- a) Stack b) Macro c) Micro d) Union.

Explanation: A micro flowchart provides a detailed, step-by-step representation of a process or algorithm, showing every individual operation and decision. It's often used to represent the internal logic of a system or program.

Answer: (c)

3. Flowcharts and algorithms are used for -

- a) Better programming
b) Easy testing and debugging
c) Efficient coding
d) All of the above

Explanation: Flowcharts and algorithms help in better programming by providing a clear structure of the logic. They make testing and debugging easier and lead to efficient coding by reducing errors and improving code quality.

Answer: (d)

4. The operation represented by parallelogram -

- a) Input/output
b) Assignment
c) Comparison
d) Condition

Explanation: In a flowchart, a parallelogram represents input/output operations, such as receiving input from a user or displaying results.

Answer: (a)

5. The number of steps on algorithm should be-

- a) Finite b) Infinite c) Fixed d) Floating

Explanation: An algorithm must have a finite number of steps to ensure that it eventually produces a result and terminates. If an algorithm had an infinite number of steps, it would never complete its task.

Answer: (a)

6. Flowcharts and Algorithms are used for- (a) better programming (b) easy testing and debugging (c) efficient coding (d) all of these.

Explanation: Flowcharts and algorithms help improve programming, debugging, and coding efficiency.

Answer: (d)

7. When designing a flowchart, each step in the process is depicted by a _____ and is associated with a short description.

Explanation: In a flowchart, each step is depicted by a symbol (shape) such as rectangles (processes), diamonds (decisions), and ovals (start/end).

Answer: Symbol (Shape)

8. With the help of an example, explain the use of flow chart.

Answer: A flowchart is a diagram that represents a process or algorithm using symbols. For example, a flowchart for checking if a number is even or odd would include:

- 1) Start
- 2) Input number
- 3) Check if the number is divisible by 2
- 4) If Yes → Print "Even"
- 5) If No → Print "Odd"
- 6) End

This graphical representation makes it easier to understand and debug algorithms.

9. Define algorithm with a suitable example.

Answer:

Algorithm: An algorithm is a well-defined, step-by-step procedure for solving a problem or achieving a specific goal. It's a set of instructions that can be followed to produce a desired output.

Example: Finding the Maximum of Two Numbers- Consider an algorithm that finds the larger of two numbers:

1. Input: Two numbers, say A and B.

2. Process:

Step 1: Compare A and B.

Step 2: If A is greater than B, then set the maximum to A.

Step 3: Otherwise, set the maximum to B.

3. Output: The maximum number between A and B.

This simple algorithm clearly outlines the steps needed to determine the greater number, ensuring that anyone following the procedure will consistently arrive at the correct result.

10. Why algorithm is used before any programming?

Answer: An algorithm is used before programming because it serves as a step-by-step plan to solve a problem logically and efficiently. Here are some key reasons why algorithms are important before writing a program:

- 1) **Clear Problem Understanding:** Helps in breaking down a complex problem into smaller, manageable steps.
- 2) **Efficient Solution Design:** Allows choosing the best approach to solve a problem before coding.
- 3) **Error Reduction:** Identifies mistakes early, reducing debugging efforts in the actual program.
- 4) **Code Optimization:** Ensures the program runs efficiently by selecting the best logic.
- 5) **Better Communication:** Helps programmers and teams understand the logic before implementation.
- 6) **Language Independence:** Algorithms can be written in plain language or flowcharts, making them adaptable to any programming language.
- 7) **Saves Time and Effort:** Avoids unnecessary coding trials by providing a clear roadmap.

11. Write the advantages of the flowcharts.

Answer:

Advantages of Flowcharts Flowcharts offer several benefits in problem-solving and programming. Here are some key advantages:

- 1) **Easy to Understand :** Uses graphical symbols to represent steps, making complex processes simple and clear.
- 2) **Effective Communication :** Helps programmers, designers, and stakeholders understand the logic of a system.
- 3) **Efficient Problem-Solving:** Provides a clear visualization of the process, making it easier to analyze and improve.
- 4) **Error Detection:** Helps in identifying logical errors before coding, reducing debugging efforts.
- 5) **Improves Documentation:** Acts as a reference for developers, ensuring consistency in project development.
- 6) **Better Debugging and Testing:** Makes it easier to locate and correct errors in the logic.
- 7) **Language Independent :** Can be used to plan algorithms without relying on a specific programming language.

12. Describe the use of different symbols used in any flow chart with suitable example.

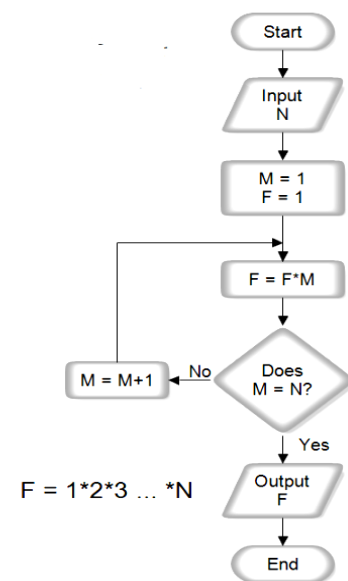
Answer:

Flowchart Symbols and Their Uses

- 1) **Oval (Terminator):** Represents the start and end of a flowchart.
 - Example: Start or End of a program.
- 2) **Rectangle (Process) :** Represents a process, operation, or action.
 - Example: Calculate Sum = A + B.
- 3) **Parallelogram (Input/Output):** Used for input (user input) or output (display results).
 - Example: Enter a number, Print the result.
- 4) **Diamond (Decision):** Represents a decision-making step (Yes/No, True/False).
 - Example: Is X > 10? (Yes → Next step, No → Another action).
- 5) **Arrow (Connector):** Indicates the direction of flow from one step to another.
 - Example: Connects all steps logically in a sequence.

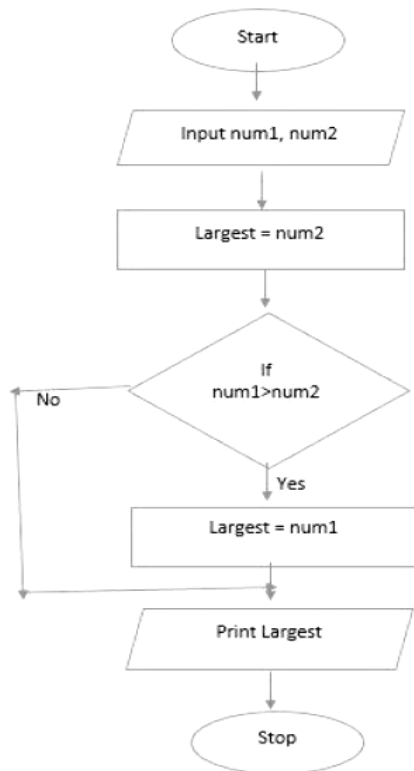
13. Draw a flow chart to calculate the factorial of an integer.

Answer: Flowchart to calculate the factorial of an integer:



14. Draw a Flowchart that will reads two values and determines the largest value and prints the largest value.

Answer: Flowchart for determining largest value of two values-



15. Algorithm & Flowchart to convert temperature from Celsius to Fahrenheit.

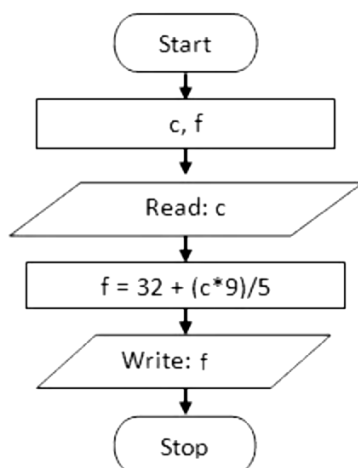
Answer: Algorithm to Convert Temperature from Celsius to Fahrenheit-

1. Start
2. Input temperature in Celsius (C)
3. Apply formula:

$$F = \left(C \times \frac{9}{5} \right) + 32$$

4. Display the temperature in Fahrenheit (F)
5. End

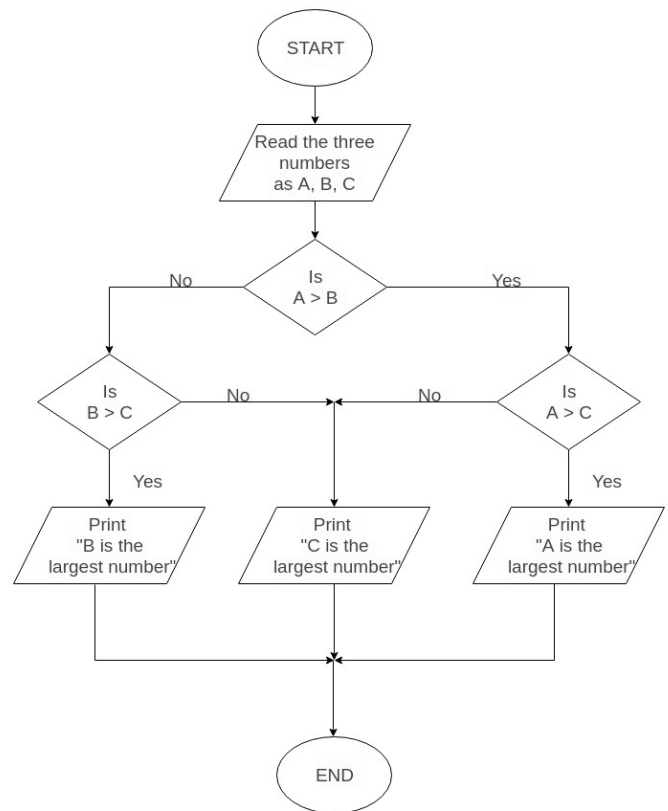
Flowchart to convert temperature from Celsius to Fahrenheit:



16. Draw a flow Chart to find out Largest number of three given number.

Answer:

Flowchart to find the largest number among three given numbers:



17. What is the difference between an algorithm and a flowchart?

Answer: An algorithm is a textual stepwise solution, while a flowchart is a graphical representation of the algorithm.

18. What are the limitations of a flowchart?

Answer: Complex logic can result in a complicated flowchart, and modifications may require redrawing the entire diagram.

19. Explain the term 'Pseudocode.'

Answer: Pseudocode is a simplified, informal, and readable version of programming logic that uses plain language to describe steps.

20. What is a recursive algorithm?

Answer: A recursive algorithm is one that calls itself to solve a smaller subproblem of the original problem.

21. What is a sequential algorithm?

Answer: A sequential algorithm follows a linear flow of control without any branching or loops.

1. **Best choice of the tools to create dynamic web page-**
a) CSS b) JavaScript c) Java d) paint.

Explanation: JavaScript is the best choice for creating dynamic web pages because it allows for real-time updates, user interactions, animations, and data manipulation without requiring a page reload.

Answer: (b)

2. **CSS stands for -**
a) Cascade style sheets
b) Color and style sheets
c) Cascading style sheets
d) None of the above

Explanation: CSS stands for Cascading Style Sheets. It is used to style and design the layout of web pages, including aspects like colors, fonts, margins, and positioning of elements. The term "cascading" refers to how styles are applied in a hierarchical manner.

Answer: (c)

3. **Best choice of the tools to introduce different style in a web page-**
a) CSS b) JavaScript c) Java d) Power point.

Explanation: CSS(Cascading Style Sheets) is the best tool for introducing different styles in a web page. It controls the layout, colors, fonts, spacing, and responsiveness of web pages, ensuring a visually appealing design.

Answer: (a)

4. **Rules to exchange data between computers are called-** (a) Procedure (b) protocols (c) hyperlink (d) program.

Explanation: A protocol is a set of rules that define how data is transmitted over networks (e.g., HTTP, TCP/IP).

Answer: (b)

5. **CPU scheduling is the basis of-** (a) Large memory size system (b) Multiprogramming system (c) Multiprocessing system (d) none of the above.

Explanation: CPU scheduling is crucial for multiprogramming, which allows multiple processes to be executed concurrently by the CPU.

Answer: (b)

6. **Which of the following errors will be handled by the operating system?** (a) Lack of paper in printer (b) Connection failure in the network (c) Power failure (d) All of the mentioned above.

Explanation: The operating system (OS) handles errors like printer paper shortages, network failures, and power failures to ensure system stability.

Answer: (d)

7. **All html tags are must be enclosed in-**
a) ? And! b) <And> c) { And } d) # And #

Explanation: In HTML (HyperText Markup Language), all tags are enclosed within angle brackets (< >). For example, <html>, <body>, and <p> are tags enclosed in angle brackets.

Answer: (b)

8. **Which of the following HTML tag is used for inserting the largest heading in HTML?** (a) <h1> (b) <h3> (c) <h5> (d) <h6>

Explanation: In HTML, <h1> represents the largest heading

Answer: (a)

9. ** tag makes the enclosed text bold. What is other tag to make text bold?** a) b) <dar> c) <black> d) <emp>

Explanation: The tag is used to make text bold and also conveys that the enclosed text is of strong importance. Unlike the tag, which only provides a visual effect, has semantic meaning, making it preferred for accessibility and SEO purposes.

Answer: (a)

10. **In how many way can CSS be written in?** (a) 1 (b) 2 (c) 3 (d) 4

Explanation: CSS can be written in three ways: Inline, Internal, and External stylesheets.

Answer: (c)

11. **When a user views a page containing a JavaScript program, which machine actually executes the scripts?** (a) The user machine running a web browser (b) The web server (c) A central machine deep within Netscape office (d) none of the above

Explanation: JavaScript is executed on the client-side (i.e., the user's web browser) rather than on a web server.

Answer: (a)

12. If we want define style for a unique element, then which CSS selector will we use? a) Id b) text c) class d) name.

Explanation: In CSS, the `id` selector is used to define a style for a unique element. Since `id` values must be unique within a page, using `#idname` allows you to apply specific styles to that particular element.

Answer: (a)

13. Which of the following CSS property is used to set the blend mode for each background layer of an element?

- a) background-blend-mode property
- b) background-collapse property
- c) background-transform property
- d) background-origin property

Explanation: The `background-blend-mode` property in CSS is used to set how the background images and colors should blend with each other and with the element's background. It applies blending effects similar to those found in image editing software, allowing for creative visual effects.

Answer: (a)

14. Inside which HTML element do we put the JavaScript? a) `<js>` b) `<script>` c) `<scripting>` d) `<javascript>`

Explanation: In HTML, the `<script>` element is used to embed or reference JavaScript code. It can be placed in the `<head>` or `<body>` section of the document.

Answer: (b)

15. Tags and text that are not directly displayed on the page are written in section- a) `<html>` b) `<head>` c) `<title>` d) `<body>`

Explanation: The `<head>` section contains metadata and elements like `<title>`, `<meta>`, `<link>`, `<script>`, and `<style>`, which are not directly displayed on the web page. The content inside the `<head>` helps in setting up the page's structure, appearance, and behavior.

Answer: (b)

16. What will the following code return: `Boolean(10>9)`
-a) NaN b) true c) false

Explanation: The expression `Boolean(10 > 9)` returns `true`. This is because the comparison `10 > 9` evaluates to `true`, and the `Boolean()` function converts the result into a boolean value.

Answer: (b)

17. A _____ language is computer language that uses tags to define elements within a document.

Explanation: A markup language (e.g., HTML, XML) uses tags to define elements within a document.

Answer: markup

18. Number of types of errors in JavaScript's is _____.

Explanation: JavaScript has six types of errors:

- 1) Syntax Errors
- 2) Runtime Errors
- 3) Logical Errors
- 4) Load-time Errors
- 5) Interface Errors
- 6) Communication Errors

Answer: 6

19. What is CSS?

Answer: CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation and layout of web pages. It allows developers to style HTML elements by defining properties such as colors, fonts, spacing, and positioning. CSS enhances the visual appeal of a website and ensures a consistent design across multiple pages.

20. Does all HTML tags have an end tag?

Answer: No, not all HTML tags have an end tag. Some tags, called self-closing tags or void elements, do not require a closing tag. Examples include `<br>`, `<img>`, `<input>`, `<hr>`, and `<meta>`.

21. Write the utility of a scripting language.

Answer: Scripting languages automate tasks, enhance web functionality, and improve user interaction. They are commonly used for web development (JavaScript, PHP), system administration (Bash, Python), and game development (Lua).

22. Define source code and Executable code.

Answer:

Source Code: It is the human-readable code written in a programming language before compilation or interpretation. It consists of instructions written by a programmer.

Example: `printf("Hello, World!");`

Executable Code: It is the machine-readable binary code generated after compiling the source code, which the computer can directly execute. It usually has extensions like `.exe` in Windows or `.out` in Linux.

23. How many ways you can apply CSS to an HTML document?

Answer: There are three main ways to apply CSS (Cascading Style Sheets) to an HTML document:

(1) **Inline CSS:** Applied directly within an HTML element using the `style` attribute.

(2) **Internal CSS (Embedded CSS):** Defined inside a `<style>` tag within the `<head>` section of an HTML document.

(3) **External CSS:** Stored in a separate `.css` file and linked using `<link>` inside the `<head>`.

24. Identify the ways to integrate CSS on a web page.

Answer: You can include CSS in a web page in three ways: inline, internal (using a `<style>` tag in the `<head>`), and external (linking to a separate `.css` file).

1. Inline CSS: Applies styles directly within HTML tags using the `style` attribute.

Example: `<p style="color:blue;">This is a paragraph.</p>`

2. Internal CSS: Defines styles within the `<head>` section of an HTML document using a `<style>` tag.

Example:

```
<head>
  <style>
    p {
      color: blue;
    }
  </style>
</head>
```

3. External CSS: Creates a separate CSS file (e.g., `style.css`) and links it to the HTML document using a `<link>` tag.

Example:

```
<head>
  <link
rel="stylesheet"
href="style.css">
</head>
```

25. Explain the differences between `<picture>` tag and `<img>` tag used in HTML.

Answer:

Differences Between `<picture>` and `<img>` Tag in HTML

<code><picture></code> Tag	<code></code> Tag
Provides a responsive image solution by allowing multiple sources.	Displays a single image from a specified URL.
Supports different image sources based on screen size or device type.	Displays the same image regardless of device.
Contains multiple <code><source></code> elements and an <code></code> fallback.	Uses only the <code></code> tag with the <code>src</code> attribute.
Ideal for responsive designs where different images are needed for different screens.	Best for simple image display.

26. Explain the utility of JavaScript.

Answer: Utilities of JavaScript-

(1) **Adds Interactivity:** Enables dynamic features such as form validation, dropdown menus, sliders, and pop-ups.

(2) **Enables Client-Side Functionality:** Runs directly in the browser, reducing server load and improving performance.

(3) **Facilitates Web Development:** Works with HTML and CSS to create modern, user-friendly websites.

(4) **Enhances User Experience:** Allows animations, transitions, and real-time content updates.

(5) **Supports Game and App Development:** Helps build browser-based games and hybrid mobile applications.

27. What are Style Sheets ?

Answer: Style Sheets (CSS - Cascading Style Sheets) are used to control the appearance and layout of web pages. They allow web developers to define styles such as colors, fonts, spacing, and positioning separately from HTML content, making web pages more visually appealing and easier to maintain.

28. What are the key difference between java and JavaScript?

Answer:

Key Differences Between Java and JavaScript:

Java	JavaScript
Object-oriented programming language (OOP).	Scripting language for web development.
Runs in a Java Virtual Machine (JVM).	Runs in a web browser or Node.js.
Compiled into bytecode before execution.	Interpreted and executed line by line.
Strict, strongly typed with explicit data types.	Loosely typed with dynamic type assignment.
Used for standalone applications, enterprise software, mobile apps (Android).	Primarily used for web development and interactive web pages.
Uses multithreading for parallel execution.	Uses asynchronous programming (event-driven model).
Platform-independent but requires JVM.	Natively runs in browsers without additional setup.
Class-based, requires object instantiation.	Prototype-based, objects can be created dynamically.

29. Differentiate between an ordered list and an unordered list used in HTML with proper example.

Answer:

Difference Between Ordered List and Unordered List in HTML

Ordered List ()	Unordered List ()
Displays items in a specific sequence or ranking.	Displays items without any specific order.
Uses numbers (1, 2, 3), letters (A, B, C), or Roman numerals (I, II, III).	Uses bullet points (•, °, ▪).
Best for steps, rankings, instructions, or priorities.	Suitable for general lists like navigation menus or feature lists.

An ordered list () is used when sequence matters, while an unordered list () is for items without a specific order.

Example of Ordered List ()

<h3>Steps to Make Tea:</h3>

Boil water.

Add tea leaves or a tea bag.

Let it steep for 5 minutes.

Strain and serve hot.

Output:

1. Boil water.
2. Add tea leaves or a tea bag.
3. Let it steep for 5 minutes.
4. Strain and serve hot.

Example of Unordered List ()

<h3>Grocery List:</h3>

Milk

Eggs

Bread

Butter

Output:

- Milk
- Eggs
- Bread
- Butter

30. Write down the various formatting tags in HTML.

Answer:

Various Formatting Tags in HTML: HTML provides several text formatting tags to style and structure content. Here are some important ones:

Tag	Description
	Bold text
	Important text (bold)
<i>	<i>Italic text</i>
	<i>Emphasized text (italic)</i>
<u>	Underlined text
<mark>	Highlighted text
<small>	Smaller text
	Strikethrough text
<ins>	Inserted text (underlined)
<sup>	Superscript text (for exponents)
<sub>	Subscript text (for chemical formulas)
<code>	Monospace (code) text
<pre>	Preformatted text (preserves spacing)
<blockquote>	Block quotation
<abbr>	Abbreviation (with tooltip)

31. Write an HTML code to display in Web browser following table:

MATH	PHYSICS	IT
80	90	95
65	78	65
89	76	70

Answer:

HTML code to display the given table in a web browser:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-scale=1.0">
  <title>Student Marks Table</title>
  <style>
    table {
      width: 50%;
      border-collapse: collapse;
      text-align: center;
      font-size: 18px;
    }
    th, td {
      border: 2px solid black;
      padding: 10px;
    }
    th {
      background-color: #f2f2f2;
    }
  </style>
</head>
<body>
```

```
<h2>Student Marks Table</h2>
```

```
<table>
  <tr>
    <th>MATH</th>
    <th>PHYSICS</th>
    <th>IT</th>
  </tr>
  <tr>
    <td>80</td>
    <td>90</td>
    <td>95</td>
  </tr>
  <tr>
    <td>65</td>
    <td>78</td>
    <td>65</td>
```

```
</tr>
<tr>
  <td>89</td>
  <td>76</td>
  <td>70</td>
</tr>
</table>
```

```
</body>
</html>
```

32. Distinguish between loop and branching statements.

Answer:

Loop Statements	Branching Statements
Loops execute a set of statements repeatedly based on a condition.	Branching statements alter the flow of execution by making decisions or jumping to another part of the program.
Used for iteration (repeating a task multiple times).	Used for decision-making or controlling program flow.
Repeats a block of code until a condition is met.	Jumps or changes execution based on conditions.
Types: for, while, do-while	Types: if, if-else, switch, break, continue, goto, return
Example (For Loop in C++): for(int i=0; i<5; i++) { cout << i; } (Prints numbers 0 to 4 repeatedly)	Example (If-Else in C++): if (x > 0) { cout << "Positive"; } else { cout << "Negative"; } (Executes one block based on the condition)

33. What is the difference between the operators '=' & '==='?

Answer:

== (Loose Equality)	=== (Strict Equality)
Converts data types if needed	No type conversion
Compares values only	Compares both values and types
Slightly slower due to conversion	Faster, no conversion needed
May produce unexpected results	Safer and more predictable
Useful in rare cases with type coercion	Preferred for most cases

34. What is the difference between static and dynamic web pages?

Answer:

Difference Between Static and Dynamic Web Pages

Static Web Page	Dynamic Web Page
Fixed content that doesn't change without manual updates.	Content changes based on user input or server response.
Created using HTML, CSS, and sometimes JavaScript.	Uses server-side scripting languages like PHP, ASP.NET, or Node.js.
Limited interactivity.	Highly interactive and responsive.
No database connection.	Often connected to a database for real-time content.
Faster as no server-side processing is required.	May be slower due to server-side processing.
Example: Personal portfolio, informational websites.	Example: E-commerce sites, social media platforms.
Harder to maintain for large websites.	Easier to manage with dynamic content management.

35. Name some CSS style components.

Answer: Here are some common CSS style components:

- 1. Color:** Defines the color of text, backgrounds, borders, etc.
- 2. Font:** Controls font family, size, style, weight, and variant.
- 3. Background:** Specifies background color, image, position, repeat, and attachment.
- 4. Border:** Defines border width, style, color, and radius.
- 5. Margin:** Sets the space between elements.
- 6. Padding:** Sets the space between content and borders.
- 7. Width and Height:** Specify the dimensions of elements.
- 8. Position:** Defines the position of elements (static, relative, absolute, fixed).
- 9. Display:** Specifies how elements are displayed (block, inline, inline-block, etc.).
- 10. Text:** Controls text alignment, decoration, transform, and other text-related properties.

36. Design a HTML page with Java Script to calculate sum of n natural numbers.

Answer: Here's a simple HTML page with JavaScript to calculate the sum of the first **n** natural numbers.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sum of N Natural Numbers</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin: 50px; }
    .container { max-width: 400px; margin: auto; padding: 20px; border: 1px solid #ddd; border-radius: 10px; }
  </style>
</head>
<body>
  <div class="container">
    <h2>Sum of N Natural Numbers</h2>
    <input type="number" id="number" min="1" placeholder="Enter n">
    <button
      onclick="calculateSum()">Calculate</button>
    <h3 id="result"></h3>
  </div>
  <script>
    function calculateSum() {
      let n =
parseInt(document.getElementById("number").value);
      document.getElementById("result").innerText = n > 0 ? `Sum: ${n * (n + 1) / 2}` : "Enter a valid number";
    }
  </script>
</body>
</html>
```

37. Define cascading.

Answer: Cascading in CSS refers to the priority order in which multiple styles are applied to an HTML element. The styles are applied based on three main factors:

- 1) Inline Styles** (highest priority) – Defined directly in an HTML tag using the `style` attribute.
- 2) Internal CSS** – Defined within a `<style>` tag inside the `<head>` section of an HTML document.
- 3) External CSS** – Linked via an external `.css` file using the `<link>` tag.

When multiple styles conflict, the browser follows the "cascade" rule to determine which style takes precedence.

38. How an image can be inserted using suitable tags of HTML in any web page?

Answer:

Inserting an Image in an HTML Web Page:

To insert an image in an HTML web page, the tag is used. This tag does not require a closing tag. Below is the basic syntax:

```

```

Explanation of Attributes:

- 1) **src (Source):** Specifies the path or URL of the image file.
 - Example: src="image.jpg" (for local images)
 - Example: src="https://example.com/image.jpg" (for online images)
- 2) **alt (Alternative Text):** Provides a description of the image in case it does not load.
- 3) **width and height:** Defines the dimensions of the image (in pixels).
- 4) **title (Optional):** Displays a tooltip when the user hovers over the image.

39. Write proper syntax to introduce Style Sheet in HTML.

Answer:

Proper Syntax to Introduce a Style Sheet in HTML

1. Inline CSS (Applied directly to an element)

```
<p style="color: blue; font-size: 18px;">This is a
styled paragraph.</p>
```

2. Internal CSS (Defined inside <style> in the <head> section)

```
<head>
  <style>
    p {
      color: blue;
      font-size: 18px;
    }
  </style>
</head>
```

3. External CSS (Linked through a separate CSS file)

```
<head>
  <link rel="stylesheet"
  href="styles.css">
</head>
```

40. Write a program to create HTML table with the following output:

Name	Department	Mobile Number
Rahul Das	DME	9123443518
Sanjib Paul	DCE	8513546728
Sibnath Dey	DSE	9175354578

Answer:

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Table</title>
  <style>
    table {
      width: 50%;
      border-collapse: collapse;
      margin: 20px 0;
    }
    th, td {
      border: 1px solid #000;
      padding: 10px;
      text-align: center;
    }
    th {
      background-color: #f2f2f2;
    }
  </style>
</head>
<body>
  <h2>Employee Details</h2>
  <table>
    <tr>
      <th>Name</th>
      <th>Department</th>
      <th>Mobile Number</th>
    </tr>
    <tr>
      <td>Rahul Das</td>
      <td>DME</td>
      <td>9123443518</td>
    </tr>
    <tr>
      <td>Sanjib Paul</td>
      <td>DCE</td>
      <td>8513546728</td>
    </tr>
    <tr>
      <td>Sibnath Dey</td>
      <td>DSE</td>
      <td>9175354578</td>
    </tr>
  </table>
</body>
</html>
```

41. Why do we use JavaScript?

Answer:

Reasons to Use JavaScript:

- 1) **Enhances Interactivity:** Enables interactive features like buttons, forms, sliders, pop-ups, and animations.
- 2) **Validates User Input:** Checks form inputs before submitting data to the server (e.g., verifying an email format).
- 3) **Handles Dynamic Content:** Updates content on a webpage without reloading (e.g., live search, real-time notifications).
- 4) **Works with HTML & CSS:** Controls and manipulates HTML elements and CSS styles dynamically.
- 5) **Client-Side Execution:** Runs directly in the user's browser, reducing server load and improving speed.
- 6) **Game & App Development** – Used to develop web-based games and applications using libraries like Phaser.js.
- 7) **Cross-Browser Compatibility:** Works on almost all modern web browsers without additional software.

42. Write a JavaScript Program to Find HCF or GCD.

Answer:

```
// JavaScript Program to Find HCF or GCD Using Euclidean Algorithm
```

```
function findHCF(a, b) {
  // Ensure both numbers are positive
  a = Math.abs(a);
  b = Math.abs(b);
```

```
  // Apply Euclidean algorithm
  while (b !== 0) {
    let temp = b;
    b = a % b;
    a = temp;
  }
  return a;
}
```

```
// Example Usage
const num1 = 72;
const num2 = 120;
console.log(`HCF of ${num1} and ${num2} is: ${findHCF(num1, num2)}`);
```

43. In how many ways a JavaScript code can be involved in an HTML file?

Answer: You can involve JavaScript in an HTML file in three main ways:

- 1) **Internal JavaScript:** JavaScript code is written directly inside a `<script>` tag within the HTML file.

```
<html>
<head>
  <script>
    console.log("Hello from internal JavaScript!");
  </script>
</head>
<body>
  <h1>Internal JS Example</h1>
</body>
</html>
```

- 2) **External JavaScript:** JavaScript code is written in a separate .js file and linked to the HTML using the `<script>` tag with the `src` attribute.

```
<html>
<head>
  <script src="script.js"></script>
</head>
<body>
  <h1>External JS Example</h1>
</body>
</html>
```

- 3) **Inline JavaScript:** JavaScript can be added directly within HTML elements using event attributes like `onclick`, `onmouseover`, etc.

```
<html>
<body>
  <h1>Inline JS Example</h1>
  <button onclick="alert('Hello from inline JavaScript!')">Click Me</button>
</body>
</html>
```

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1. A small program that gets loaded in the computer without the user knowledge and replicate itself repeatedly called: (a) Worms (b) Virus (c) Browser (d) None of these.

Explanation: A worm is a self-replicating program that spreads without user intervention, unlike viruses that require execution of an infected file.

Answer: (a)

2. What is the term for unsolicited email? (a) Spam (b) Worm (c) Browser (d) News Group.

Explanation: Spam refers to unwanted, bulk emails that may contain advertisements, phishing attempts, or malware.

Answer: (a)

3. Which of the following is not a cybercrime? (a) Denial of Service (b) Man in the middle (c) Malware (d) AES.

Explanation: AES (Advanced Encryption Standard) is an encryption algorithm, not a cybercrime.

Answer: (d)

4. In ethical hacking and cyber security, there are ____ types of scanning. a) 1 b) 2 c) 3 d) 4

Explanation: In ethical hacking and cybersecurity, there are three types of scanning:

- 1) **Port Scanning:** Identifies open ports and services running on a target system.
- 2) **Network Scanning:** Discovers devices connected to a network and gathers information about them.
- 3) **Vulnerability Scanning:** Detects vulnerabilities or weaknesses in a system that could be exploited by attackers.

5. Which of the following is a programs that copy themselves throughout a computer or network?

a) Worms b) Trojans c) Viruses d) Rootkits

Explanation: Worms are malicious programs that can replicate and spread themselves across computers or networks without any human intervention. Unlike viruses, which require a host file to attach to, worms operate independently, often exploiting vulnerabilities in network security to spread rapidly.

Answer: (a)

6. Which of the following port and IP address scanner famous among the users? a) Cain and Abel b) Angry IP Scanner c) Snort d) Ettercap

Explanation: Angry IP Scanner is a popular and widely used IP address and port scanner. It is known for its speed and simplicity, making it a favorite among network administrators and cybersecurity professionals for scanning and managing networks.

Answer: (b)

7. Which of the following transmission directions listed is not a legitimate channel? a) Simplex b) Half Duplex c) Full Duplex d) Double Duplex

Explanation: Double Duplex is not a legitimate transmission mode. The three valid types of data transmission are:

- 1) **Simplex:** Data flows in one direction only (e.g., TV broadcasting).
- 2) **Half Duplex:** Data flows in both directions, but only one direction at a time (e.g., walkie-talkies).
- 3) **Full Duplex:** Data flows in both directions simultaneously (e.g., telephone calls).

Answer: (d)

8. ____ attack is an attempt to make a computer resource unavailable to its intended.

Explanation: A Denial-of-Service (DoS) attack aims to make a computer resource unavailable by overwhelming it with excessive requests.

Answer: Denial-of-Service (DoS)

9. How does a virus get installed in a computer?

Answer: A virus can enter a computer through infected email attachments, malicious websites, software downloads, USB drives, or social engineering attacks. Once installed, it can replicate, corrupt files, steal data, or harm system performance.

10. Is worm a computer virus? Write your opinion.

Answer: No, a worm is not a computer virus, but both are types of malware. A worm is a self-replicating program that spreads automatically across networks without needing a host file or user action. It spreads through vulnerabilities, emails, or the internet.

In conclusion, worms and viruses are different, but both require antivirus protection to prevent infections.

11. What is cyber stalking? What is Spam?

Answer:

Cyber Stalking: Cyber stalking is the use of the internet, social media, emails, or other digital communication to harass, intimidate, or threaten someone. It involves repeated and unwanted attention, such as sending threatening messages, spreading false information, tracking online activities, or hacking into personal accounts. Cyber stalkers may target individuals for personal revenge, obsession, or financial gain, causing emotional distress, fear, or even real-world harm. Laws against cyber stalking exist in many countries to protect victims and penalize offenders.

Spam: Spam refers to unwanted, irrelevant, or unsolicited messages sent over the internet, typically in bulk. It is commonly found in emails, messages, social media, and advertisements. Spammers use automated bots to send these messages, often for promotional, fraudulent, or malicious purposes.

12. What is computer hacking?

Answer:

Computer Hacking: Computer hacking is the act of gaining unauthorized access to computer systems, networks, or data, often by exploiting security vulnerabilities. Hackers may do this for various reasons, including stealing sensitive information, disrupting operations, or testing security systems. While some hacking is illegal and harmful (such as cybercrime, identity theft, or malware attacks), ethical hacking is performed by cybersecurity professionals to identify and fix security flaws. Effective cybersecurity measures help prevent hacking and protect digital assets from unauthorized access.

13. Write the advantages of Ethical Hacking.

Answer:

Advantages of Ethical Hacking:

- 1) **Enhances Cybersecurity:** Identifies and fixes security vulnerabilities before cybercriminals exploit them.
- 2) **Prevents Data Breaches:** Protects sensitive information from hackers, reducing the risk of data leaks.
- 3) **Improves System Protection:** Strengthens firewalls, encryption, and security protocols to defend against attacks.

- 4) **Identifies Weaknesses Proactively:** Detects loopholes in networks, software, and hardware before they are exploited.
- 5) **Increases Customer Trust:** Secure systems boost confidence among clients and users.

14. Why computer security is important?

Answer: Computer security is important because-

- (1) It protects personal and financial data, prevents cyber attacks, ensures privacy, and maintains system integrity.
- (2) It safeguards sensitive information from hackers, malware, and unauthorized access, reducing the risk of identity theft, fraud, and financial loss.
- (3) Strong security measures also help businesses maintain customer trust, comply with legal regulations, and prevent disruptions caused by cyber threats.
- (4) By ensuring data confidentiality, integrity, and availability, computer security plays a crucial role in protecting individuals, organizations, and governments in the digital world.

15. Can you protect your personal computer from being hacked? How?

Answer: Yes, you can protect your personal computer from being hacked by following these essential security measures:

- 1) **Use Strong Passwords:** Create complex, unique passwords and enable multi-factor authentication (MFA) for extra security.
- 2) **Keep Software Updated:** Regularly update your operating system, antivirus, and applications to patch security vulnerabilities.
- 3) **Install Antivirus and Firewall:** Use trusted security software and enable firewalls to block unauthorized access.
- 4) **Avoid Suspicious Links and Downloads:** Do not click on unknown links or download files from untrusted sources to prevent malware infections.
- 5) **Regularly Back Up Data** – Keep backups of important files on external drives or cloud storage to prevent data loss from attacks.
- 6) **Disable Unnecessary Services:** Turn off unused Bluetooth, remote access, and other services that hackers can exploit.

16. What is a Malware and types of Malwares?

Answer:

Malware: Malware (short for malicious software) is any software or program designed to disrupt, damage, or gain unauthorized access to a computer system, network, or data. Hackers and cybercriminals use malware to steal information, harm devices, or spy on users.

Types of Malware

- 1) **Virus:** Attaches itself to a legitimate program and spreads when the program runs.
- 2) **Worm:** A self-replicating malware that spreads without user action.
- 3) **Trojan Horse:** Disguised as a legitimate program but carries harmful code.
- 4) **Ransomware:** Encrypts files and demands payment for decryption.
- 5) **Spyware:** Secretly monitors user activity and collects data.
- 6) **Adware:** Displays unwanted ads and may track browsing habits.

17. What is DoS Attack? What are the security measures against DoS attack?

Answer:

DoS Attack: A Denial-of-Service (DoS) attack is a cyberattack where attackers overload a system, network, or server with excessive requests, making it slow or unavailable to legitimate users. The goal is to disrupt services, cause downtime, or crash the system.

Security Measures Against DoS Attacks:

- 1) **Firewalls & Intrusion Detection Systems (IDS) :** Use advanced firewalls and IDS/IPS to detect and block malicious traffic.
- 2) **Rate Limiting:** Restrict the number of requests a user can make in a short time.
- 3) **Load Balancers** – Distribute traffic across multiple servers to prevent overloading.
- 4) **DDoS Protection Services:** Use cloud-based services like Cloudflare, AWS Shield, or Akamai to absorb attacks.
- 5) **IP Blacklisting & Whitelisting:** Block malicious IPs and allow only trusted ones.
- 6) **Traffic Filtering:** Filter abnormal traffic patterns using AI-based security tools.
- 7) **Content Delivery Network (CDN):** A CDN helps distribute traffic and absorb attack loads.
- 8) **Network Redundancy:** Deploy backup servers to handle sudden traffic spikes.
- 9) **Regular Security Audits:** Identify vulnerabilities before attackers exploit them.
- 10) **Incident Response Plan:** Prepare a response strategy to mitigate the impact of an attack.

18. How does a computer worm work?

Answer:

Working of a Computer Worm:

- 1) **Infiltration:** The worm enters a system through email attachments, software vulnerabilities, or malicious links.
- 2) **Replication:** It makes multiple copies of itself and spreads across networks or external drives.
- 3) **Propagation:** The worm moves from one device to another through shared files, emails, or the internet.
- 4) **Execution of Payload:** Some worms carry harmful code that can delete files, steal data, or slow down systems.
- 5) **Persistence:** The worm remains in the system, often hiding in system files, to continue spreading or executing malicious tasks.

19. What do you mean by the term 'Ethical Hacking'?

Answer:

Ethical Hacking: Ethical hacking refers to the legal and authorized practice of testing computer systems, networks, or applications to identify and fix security vulnerabilities. Ethical hackers, also known as white-hat hackers, use hacking techniques to strengthen cybersecurity and protect organizations from cyber threats.

Purpose of Ethical Hacking:

- 1) **Identify Security Weaknesses:** Detect vulnerabilities in systems before malicious hackers exploit them.
- 2) **Prevent Cyber Attacks:** Protect data from hacking, viruses, and malware.
- 3) **Improve System Security:** Help organizations strengthen firewalls, encryption, and security protocols.
- 4) **Ensure Compliance:** Assist companies in meeting cybersecurity regulations and industry standards.

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20. Write the differences between Phishing and Spamming.

Answer:

Differences Between Phishing and Spamming:

Phishing	Spamming
A fraudulent attack where hackers trick users into providing sensitive information like passwords, credit card details, or personal data.	The act of sending unwanted bulk messages (emails, texts, or ads) to multiple recipients without their consent.
Designed to steal personal data or financial information by pretending to be a legitimate entity (e.g., fake banking emails).	Mainly used for advertising, promotions, or spreading malware, often without targeting specific victims.
Uses deceptive emails, fake websites, or messages that mimic trusted sources (e.g., banks, social media sites).	Sends irrelevant or excessive emails to a large audience, often causing annoyance.
Highly dangerous, as it can lead to identity theft, financial fraud, or malware installation.	Mostly harmless but can slow down email servers and may sometimes carry malicious links.
Example: A fake email pretending to be from a bank asking for login credentials.	Example: Receiving repeated promotional emails from unknown companies.

21. Why Firewalls have become so important today?

Answer:

Reasons for the Importance of Firewalls:

- 1) **Prevents Unauthorized Access:** Blocks hackers and unauthorized users from accessing private networks.
- 2) **Protects Against Malware & Viruses:** Filters out malicious traffic, preventing malware, ransomware, and viruses from infecting systems.
- 3) **Monitors Network Traffic:** Analyzes incoming and outgoing data to detect suspicious activity.
- 4) **Enhances Data Security:** Prevents data breaches by blocking unauthorized attempts to access sensitive information.
- 5) **Ensures Safe Online Browsing:** Restricts access to harmful websites that may contain phishing scams or malicious software.
- 6) **Reduces Risk of Cyber Attacks:** Acts as the first line of defense against hacking attempts, DoS (Denial of Service) attacks, and other threats.
- 7) **Essential for Businesses:** Protects company data, customer information, and financial transactions from cybercriminals.

22. List some properties of virus.

Answer:

Properties of a Virus

- 1) **Self-Replication:** Can copy itself and spread to other programs or files.
- 2) **Infection:** Attaches to executable files, documents, or boot sectors.
- 3) **Destructiveness:** May corrupt or delete data, crash systems, or damage software.
- 4) **Stealth:** Some viruses hide their presence using techniques like encryption or code obfuscation.
- 5) **Triggering Mechanism:** Activates at a specific time, date, or event.
- 6) **Payload:** Performs malicious actions like data theft or system damage.
- 7) **Host Dependence:** Requires a host program or file to execute and spread.

23. How does a virus get installed in a computer?

Answer:

Common Ways a Virus Gets Installed:

- 1) **Infected Email Attachments:** Opening malicious attachments in phishing emails can install viruses.
- 2) **Downloading Suspicious Files:** Downloading software, games, or media from untrusted websites may introduce viruses.
- 3) **Clicking on Malicious Links:** Clicking on pop-ups or fake ads can trigger virus downloads.
- 4) **USB Drives & External Devices:** Inserting an infected USB or external hard drive can transfer viruses.
- 5) **Fake Software Updates:** Malware disguised as system updates can install viruses unknowingly.
- 6) **Cracked Software & Torrents:** Downloading pirated or cracked software often comes with hidden viruses.
- 7) **Drive-by Downloads:** Some websites automatically download and install viruses without user consent.
- 8) **Network Infections:** If one device in a network is infected, the virus can spread to other connected systems.

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